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Multidimensional Quality Characteristics of Sichuan South-Road Dark Tea and Its Chemical Prediction

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Abstract: The distinctive quality of Sichuan south-road dark tea (SSDT) is gradually disappearing with processing innovation. Here, near-infrared (NIR) spectroscopy (NIRS) and spectrofluorometric techniques were utilized to determine the spectral characteristics of dried SSDT and its brew, respectively. Combined with chemical analysis, the multidimensional quality characteristics of SSDT will be presented. Finally, the NIR spectral fingerprint of dried SSDT was observed, with Kangzhuan (KZ) and Jinjian (JJ) showing a very similar NIR spectrum. The SiPLS models effectively predicted the levels of theabrownin, caffeine, and epigallocatechin gallate, based on the NIR spectrum, with root-mean-square errors of calibration of 0.15, 0.12, and 0.02 for each chemical compound, root-mean-square errors of prediction of 0.20, 0.09, and 0.03, and both corrected and predicted correlation coefficients greater than 0.90. Meanwhile, the fluorescence characteristics of the SSDT brew were identified based on the parallel factor analysis for the fluorescence excitation–emission matrix (EEM). The KZ and JJ brews could be classified with 100% accuracy using extreme-gradient-boosting discriminant analysis. The integration of NIRS and fluorometric EEM seems to be a powerful technique for characterizing SSDTs, and the results can greatly benefit the production and quality control of SSDTs.

Keywords: Sichuan south-road dark tea; near-infrared spectroscopy; fluorescence excitation-emission matrix; extreme-gradient-boosting discriminant analysis

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Citation: Zou, Y.; Li, X.; Han, D. Multidimensional Quality Characteristics of Sichuan South-Road Dark Tea and Its Chemical Prediction. *Agronomy* **2024**, *14*, 1582. https://doi.org/10.3390/agronomy14071582

Academic Editors: Pedro Javier Zapata and Alessandra Morana

Received: 22 May 2024
Revised: 4 July 2024
Accepted: 18 July 2024
Published: 20 July 2024

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1. Introduction

Sichuan south-road dark tea (SSDT), mainly produced in the Ya'an district, Sichuan province (Figure S1), is the traditional bulk of China's dark teas. It is an important post-harvest horticultural product processed from the mature leaves and stems of the tea plant (*Camellia sinensis*). SSDT has excellent physiological activities of hypoglycemic, hypolipidemic, and greasiness-relieving effects [1], and it is usually characterized by a brown-brick appearance, pure and aged tea aroma, mellow taste, and yellowish red brew color [2]. It is popular in southwest China, with Kangzhuan (KZ) and Jinjian (JJ) as the main product types. Generally, KZ and JJ share the same manufacturing process, with only varying ratios of tea leaves and stems. Although research is ongoing to reveal the chemical characteristics and physiological functions of KZ and JJ [3], consumers still cannot distinguish between them when drinking.

Pile fermentation, a spontaneous fermentation lasting 30 to 40 days, is the key process for SSDT quality formation. However, many tea factories have started using a new short-term, high-temperature process to speed up SSDT production instead of pile fermentation. This has led to the gradual disappearance of the unique quality of traditional SSDT, such as the pure and aged tea aroma and flavor generated by microbes, and the mellow taste. Thus, exploring and recording the multidimensional fingerprints of traditional SSDT and developing an effective identification method to differentiate between KZ and JJ is particularly important for preserving the unique quality of traditional SSDT and guiding its production.

Nowadays, the near-infrared spectroscopy technique (NIRS), which can accurately reflect the chemical composition via analyzing the absorption of organic compounds in the near-infrared region, specifically targeting hydrogen-containing groups, i.e., O-H, N-H, and C-H, etc., has been widely used in food and beverage quality research [4]. It has been well applied to detect tea polyphenols in fresh tea leaves [5], predict glucose and sucrose in black tea [6], monitor carotenoid changes of matcha during the drying process [7], evaluate bitterness or astringency of Pu'erh ripe tea [8], and determine the fixation quality of green tea [9], etc. Moreover, NIRS, coupled with certain compatible algorithms, has the advantage of being a stable and accurate nondestructive testing technology (NDT).

Indeed, the characteristics of dried tea play a crucial role in categorizing different tea types, which could be the first impression of tea quality and influence people's willingness to purchase [10]. As for the tea brew, it is also a significant indicator of tea quality, as it affects the perception of tea flavor during consumption. However, to date, the characteristics of dried SSDT and its brew are still obscure due to insufficient research, resulting from the limitations in production regions or specific consumer demand for SSDT [11]. Based on the published literature, NIRS may be the most appropriate method for revealing the overall characteristics of dried SSDT, but the appropriate method to reveal the global characteristics of SSDT brew is still under exploration. Recently, the three-dimensional fluorescence technique has emerged as an effective method to investigate the fingerprint of liquid samples [12], which shows amazing prospects in wine chemical analysis [13], wine sensory attribute prediction [14], and wine authentication [15]. It has also been well utilized to discriminate matcha producer and grade [16], and brick tea brands [17]. This technique could simultaneously scan both the excitation wavelength (λex) and emission wavelength (λem) at a constant difference of $\Delta\lambda = \lambda em - \lambda ex$ [18], and form an absorbance–transmission and fluorescence excitation–emission matrix to guarantee complete information about the fluorescent species derived from the sample.

Therefore, the objective of this work is to comprehensively investigate the overall fingerprints of traditional SSDT by chemical analysis, the NIRS determination of dried tea, and three-dimensional fluorescence spectral identification of tea brew. Additionally, some NIR spectral models for chemical prediction will be established. The results may help preserve the unique quality characteristics of traditional SSDT, standardize its production, and facilitate rapid, accurate, and non-destructive testing of SSDT chemicals.

2. Materials and Methods

2.1. Samples

SSDT samples, produced by the major dark tea factories in Ya'an City, Sichuan Province, were collected from the local market. A total of 56 samples with the typical sensory quality of traditional SSDT, consisting of 31 Kangzhuan and 25 Jinjian, were selected for experiments.

2.2. Quantification of Tea Quality Chemicals

Water extracts (WEs) in SSDT were quantified according to the Chinese national standards GB/T 8305-2013 [19]. Free amino acid (AA) was determined based on the ninhydrin colorimetric method [20], and tea polyphenols (TPs) were measured by the Folin-Ciocalteu colorimetric method [21]. Furthermore, soluble sugars (SS) were quantified utilizing the anthrone-sulfuric acid colorimetric method with slight modifications [22], and the levels of catechin monomers including (−)-epigallocatechin gallate (EGCG), (−)-epigallocatechin (EGC), (−)-epicatechin gallate (ECG), (−)-epicatechin (EC), and (+)-catechin (C), as well as caffeine (Caf), were determined by high-performance liquid chromatography according to GB/T 8313-2018 [23]. Water soluble pectin (SP) was determined by water extraction and alcohol precipitation, as described by He et al. [24]. Theaflavins (TFs), thearubigins (TRs), and theabrownins (TBs) were extracted with organic solvents and determined by spectrophotometric analysis [25].

2.3. NIR Spectral Analysis of Dried SSDT

2.3.1. NIR Spectral Acquisition

A Thermo Antaris II Fourier Transform (FT) NIR spectrometer (Thermo Fisher Scientific Inc., Madison, WI, USA), coupled with an integrating sphere, was used to collect the NIR spectrum of dried SSDT. Before scanning, the instrument was fully warmed up for 1 h, and then the SSDT sample (15 ± 0.1 g) was placed in a specially designed sample cup that was rotated 360° during scanning. The NIR spectra were collected in diffuse reflectance mode at a resolution of 4 cm^{-1}; 32 scans were recorded at 3.856 cm^{-1} intervals and averaged as one spectrum for each sample. In total, 1557 spectral variables were observed within the NIR wavelength range of 4000 to 10,000 cm^{-1}. The spectral collection for each sample was conducted 3 times. The laboratory maintained a temperature of 25 °C and a relative humidity of 60% during spectral collection.

The NIR spectral signals were exported by OMNIC 9.8 software (Thermo Fisher Scientific Inc., Waltham, MA, USA) and analyzed with TQ Analyst 9.4.45 software (Thermo Fisher Scientific Inc., Waltham, MA, USA) and Matlab R2019b (MathWorks, Natick, MA, USA).

2.3.2. Pretreatment of Raw Spectra and Establishment of Prediction Models

The calibration and prediction sets were divided as follows: the samples were ranked according to their chemical levels. Then, every three samples were selected, with two samples used for the calibration set and one for the prediction set. The level range of each chemical compound in the calibration set should cover those in the prediction set. Details are shown in Table S1.

Due to the background and external noise in the raw NIR spectrum, four pretreatment methods, including standard normal variate (SNV), multiplicative scatter correction (MSC), SNV + first derivative (1st Der), and SNV + second derivative (2nd Der), were independently applied to the raw spectra to remove the undesired factors, and the optical pretreatment, which greatly improves the signal-to-noise ratio of the spectra, was selected for the further modeling.

The preprocessed spectra were equally divided into different spectral subintervals, and a combination of 3 or 4 optimal spectral subintervals was utilized to establish the synergy interval partial least squares (SiPLS) models for SSDT chemical prediction. The accuracy and robustness of the models were evaluated using the root-mean-square error of calibration (RMSEC), root-mean-square error of prediction (RMSEP), correlation coefficients of calibration (Rc), and correlation coefficients of prediction (Rp).

2.4. Analytical Procedure for Fluorescent EEM of SSDT Brew

The SSDT brew was prepared following the Chinese national standard GB/T 23776-2018 [26], with certain modifications. Each SSDT sample, weighing 3 g, was brewed with 150 mL of boiling ultrapure water for 10 min, the brew was then filtered with quantitative filter paper (Whatman, Dassel, Germany) and centrifuged at 9300× *g* for 10 min. It was subsequently diluted 30 times with a 50% aqueous ethanol solution (pH = 2), vortexed to mix well, ultrasonicated to remove air bubbles, placed in a fluorescence cuvette, and ultimately analyzed with a HORIBA Scientific Aqualog spectrophotometer (Quark Photonics, Adelaide, Australia) under the condition that the excitation wavelength range was 240–800 nm with 5 nm increments at medium gain and 0.2 s integration time, the emission wavelength range was 242–824 nm with 4.66 nm increments. EEMs and CIE 1931 parameters were recorded for each sample. Origin software (version 8.6, OriginLab Corporation, Northampton, MA, USA) was used to display the obtained data. The water Raman scattering units for the specified emission conditions were utilized to normalize the EEMs data, then the data were corrected for the influence of inner filter effects (IFEs) and Rayleigh masking before further analysis [12].

2.5. Statistical Analysis

EEM data were processed using SOLO-MIA (version 9.2 Eigenvector Research, Inc., Manson, WA, USA). Parallel factor analysis (PARAFAC) was conducted to present the characteristic fluorescence spectral signals of the SSDT brew [27]. To discriminate between KZ and JJ, extreme-gradient-boosting discriminant analysis (XGBDA) was performed, with partial least squares (PLS) compression and a maximum of 25 latent variables, as well as mean centering preprocessing and decluttering with generalized least squares weighting (GLSW) at 0.2 to both calibrate and cross-validate (k = 10, Venetian blind procedure). Principal component analysis (PCA) was also performed based on EEM data. Descriptive statistics for chemicals were carried out with SPSS 22.0 (SPSS, Inc., Chicago, IL, USA).

3. Results

3.1. Typical Quality Chemical Profile of SSDT

The typical quality chemicals of SSDT were analyzed. It was found that the levels of WEs, TPs, AA, Caf, SS, SP, and total catechin (Cat) ranged from 14.52% to 45.64%, 2.24% to 11.3%, 0.24% to 2.39%, 1.21% to 3.71%, 0.89% to 8.62%, 0.17% to 1.21%, and 0.67% to 5.76% in the SSDT samples, respectively, while the levels of EGCG, EGC, ECG, EC, and C ranged from 0.08% to 0.81%, 0.03% to 0.47%, 0.05% to 0.22%, 0.03% to 0.23%, and 0.02% to 0.16%, respectively. Simultaneously, the levels of TFs, TRs and TBs varied from 0.02% to 0.13%, 0.64% to 2.21%, and 5.08% to 9.03%, respectively, with the lowest coefficient of variation (CV) observed in TBs (14.43%) (Table 1). TBs, showing a strong hypolipidemic activity [28], seems to be the most stable compound among the SSDT samples. Conversely, the highest CV (84.61%) was detected in EGC, a colorless and biologically active compound readily oxidized by polyphenol oxidase and peroxidase [29].

Table 1. Chemical profile of SSDT.

Chemicals	Range (%)	Mean (%)	SD	CV (%)
WEs	14.52–45.64	23.51	7.09	30.15
TPs	2.24–11.30	5.04	2.17	43.05
AA	0.24–2.39	0.71	0.46	64.78
Caf	1.21–3.71	2.02	0.56	27.31
SS	0.89–8.62	2.99	1.61	53.84
SP	0.17–1.21	0.64	0.29	45.31
Cat	0.67–5.76	1.87	1.10	58.82
EGCG	0.08–0.81	0.28	0.19	67.85
EGC	0.03–0.47	0.26	0.22	84.61
ECG	0.05–0.22	0.09	0.04	45.55
EC	0.03–0.23	0.10	0.04	36.36
C	0.02–0.16	0.05	0.03	60.78
TFs	0.02–0.13	0.05	0.44	37.73
TRs	0.64–2.21	1.37	0.56	31.38
TBs	5.08–9.03	6.51	0.39	14.43

Moreover, a clustering heat map was constructed to illustrate the relationship between the SSDT samples. It was evident that KZ and JJ were thoroughly mixed and could not be clearly distinguished from each other (Figure 1), which can be attributed to their highly similar manufacturing process and raw materials. Thus, in the following spectral analysis, KZ and JJ could possibly be treated as a whole, without any distinction.

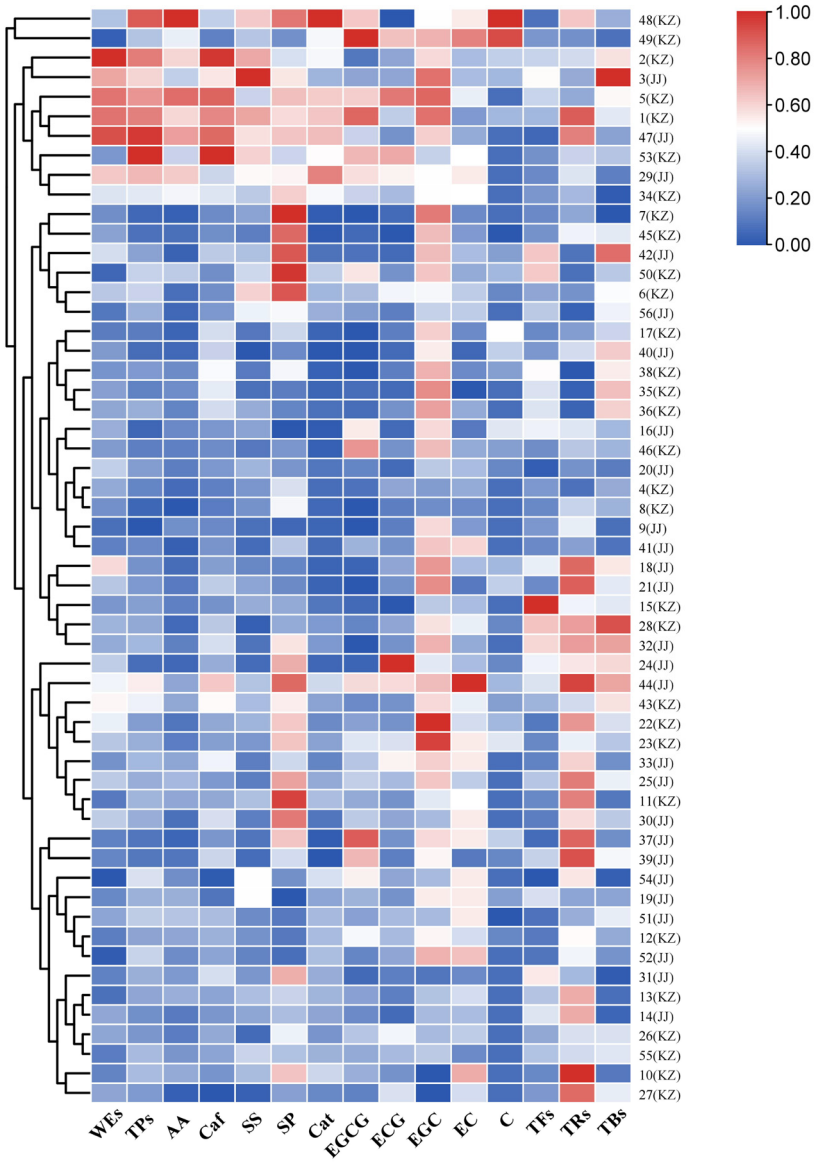


Figure 1. Hierarchical clustering heat map of tested chemicals in SSDT. The rows represent different KZ and JJ samples. Differences in chemical levels are shown by the color gradient of the blocks, and the scale bar shows the variation range of chemical normalized levels.

3.2. NIR Spectral Characteristics of Dried SSDT and the Chemical Prediction

3.2.1. Overview of NIR Spectrum

All samples tested showed a similar original spectral profile, with varying band intensities (Figure 2a). These spectral curves were relatively flat at 10,000–8500 cm^{-1}, with a small absorption peak at 8334 cm^{-1}. This peak is likely to be attributed to the second-order overtones of the C-H group stretching and may be associated with tea polyphenols [30]. Meanwhile, more absorption peaks with higher absorbance values were observed in the 8500–4000 cm^{-1} band, suggesting that rich chemical information of SSDT was reflected there. In addition, the absorption peak at 5780 cm^{-1} was mainly caused by the first overtone of C-H stretching in CH_2 and CH_3, which may be related to catechins, tea polyphenols and alkaloids [31]. The absorption peaks at 5190 cm^{-1} and 6816 cm^{-1} were mainly located in the water absorption band, corresponding to the stretching vibration of O-H and N-H, and are possibly linked to amino acids and proteins [32]. Furthermore, the absorption band between 4500–4000 cm^{-1} is associated with a combination of C-H stretching and C-H deformation, as well as the second overtone of C-H deformation. This band is probably

related to tea polyphenols and caffeine [33]. In addition, highly similar spectral curves between KZ and JJ were discovered (Euclidean distance = 0.00087), with KZ exhibiting a higher average spectral value (Figure 2b). It is suggested that KZ and JJ could be integrated to construct the sample sets for the subsequent NIR spectral model establishment, and KZ may have a higher chemical level. However, the spectra of seven samples that failed Hotelling's T2 test at a 95% confidence interval were excluded as outliers.

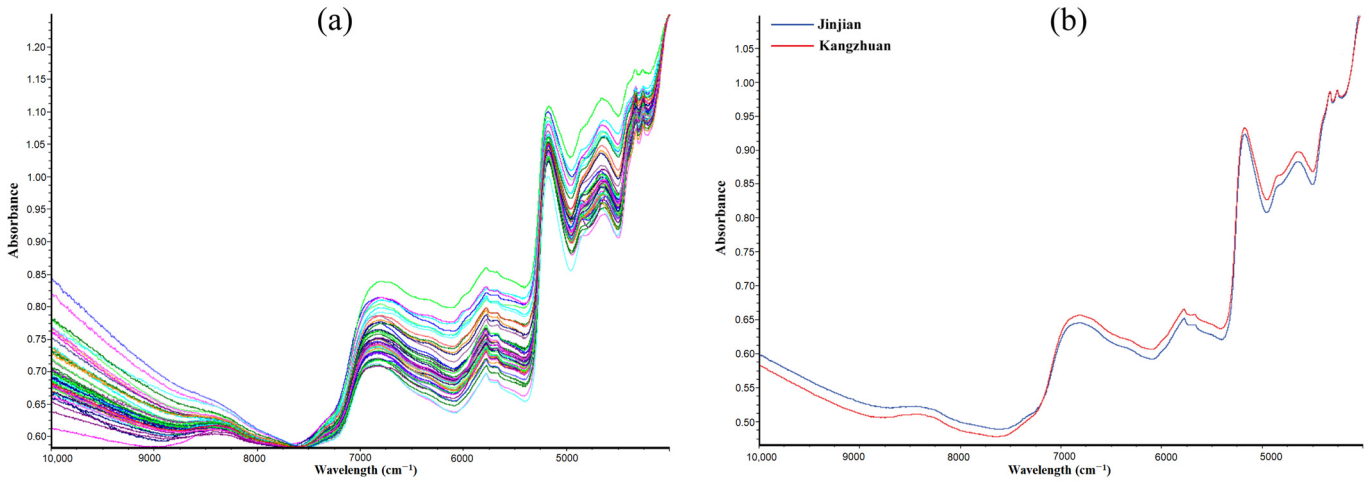


Figure 2. NIR spectral profile of dried SSDT: (**a**) original spectra of dried SSDT; (**b**) mean spectra of KZ and JJ.

3.2.2. Optimal Pretreatment of NIR Spectra

The pretreatment of raw spectra is essential, as they are susceptible to high-frequency noise, baseline drift, and light scatter. In this work, four pretreatment methods, including SNV, MSC, SNV + 1st Der, and SNV + 2nd Der, were independently evaluated in combination with PLS models, using TBs as the dependent variable due to its stability between SSDT samples. Finally, it was found that the performance of the PLS models was improved by these preprocessing methods, of which the SNV + 2nd Der pretreatment produced the best results (Rc = 0.8841; RMSEC = 0.95; Rp = 0.8306; RMSEP = 1.02) (Table S2). SSDT usually has a rough and uneven surface due to the raw materials and microbial activities during pile fermentation [34], which may lead to more scattered signals. Since SNV can effectively eliminate the influence of diffuse reflections caused by solid particle size, surface scattering, and wavelength transformations [35], its combination with the 2nd Der greatly improved the spectral signal-to-noise ratio, resulting in the best performance.

3.2.3. SiPLS Models for Chemical Prediction

Previous research indicates that the SiPLS modeling method, which splits the NIR spectrum into multiple subintervals and calculates all possible PLS models using combinations of these subintervals, can significantly improve model performance [36]. Here, it was used to establish some SSDT chemical prediction models. The preprocessed spectra were uniformly divided into 10, 15, 20, or 30 subintervals, then certain spectral subintervals were selected, and a combination of three or four optimal spectral subintervals was employed to establish predictive models. It is worth noting that the SiPLS models selected the combinations of optimal subintervals (7, 8, 12) for the TBs prediction, (3, 4, 8, 9) for the Caf prediction, and (2, 7, 10) for the EGCG prediction, which were highly effective (Table S3). Among them, the SiPLS-TBs model (RMSEC = 0.15, Rc = 0.94, RMSEP = 0.20, Rp = 0.90) was observed by dividing the full spectrum into 20 subintervals with the selected wavenumber ranges of 5796.97–6097.81, 6101.66–6398.65, and 7305.03–7602.02 cm^{-1} (Figure 3a,b). These subintervals probably reflect the information linked to the telescopic vibration of multiple phenolic hydroxyl groups [4]. Additionally, the optimal SiPLS-Caf model (RMSEC = 0.12, Rc = 0.92, RMSEP = 0.09, Rp = 0.90) was established by dividing the full spectrum into

15 subintervals with the selected wavenumbers of 4801.88–5199.14, 5203.00–5600.26, 6807.48–7204.75, and 7208.60–7605.87 cm^{-1} (Figure 3c,d). These spectral bands are mainly associated with the stretching vibrations of N-H and C-H groups and may reflect the information of caffeine [37]. Simultaneously, the SiPLS-EGCG model (RMSEC = 0.02, Rc = 0.96, RMSEP = 0.03, Rp = 0.92) achieved excellent results when the full spectrum was divided into 20 subintervals and the wavenumbers in the ranges of 4296.624–4597.464, 5796.97–6097.81, and 6703.35–7004.191 cm^{-1} were selected (Figure 3e,f). Indeed, EGCG is a major active constituent of tea polyphenols, and it has characteristic spectral subintervals related to the O-H and C-H groups in the phenolics [38].

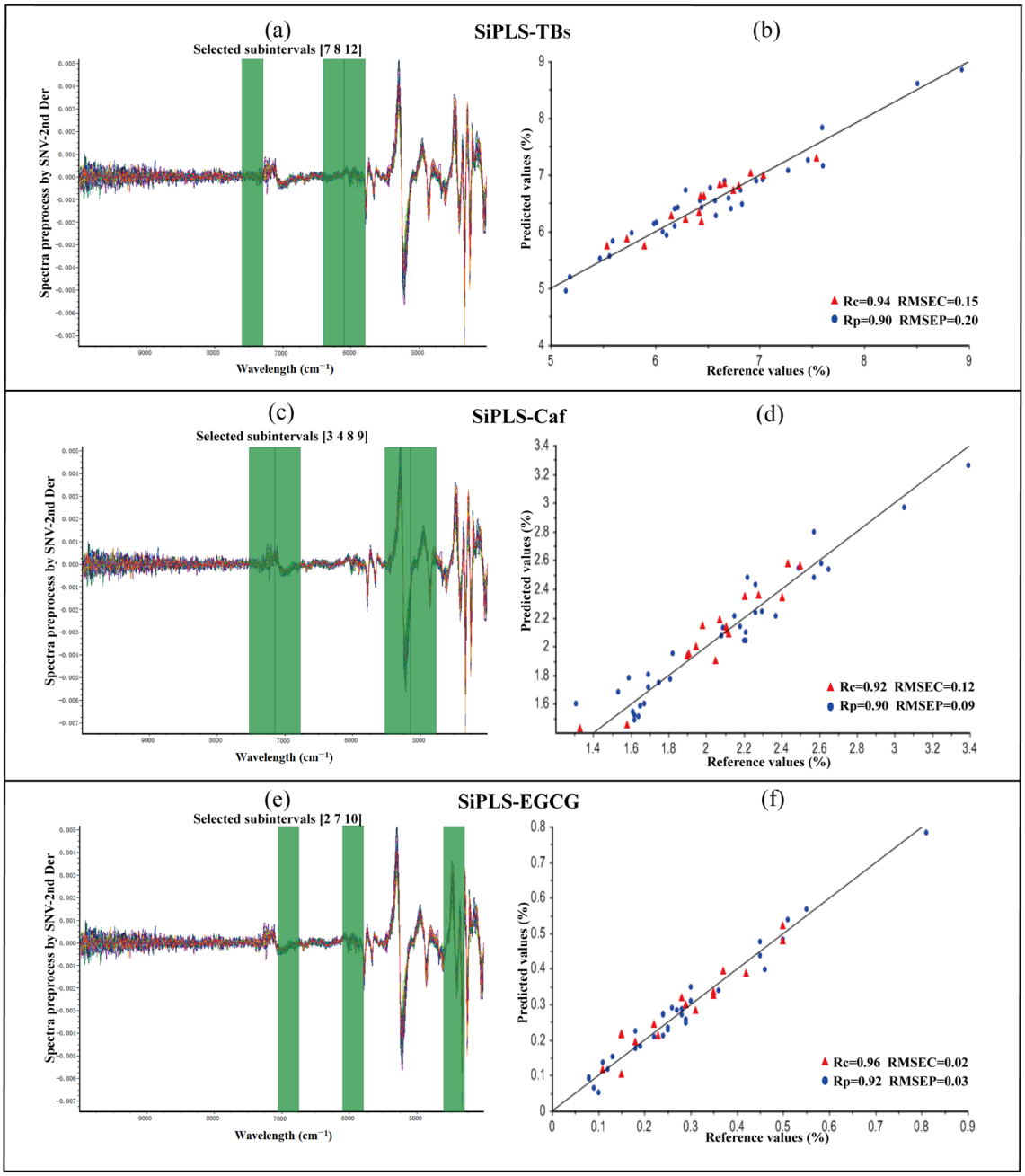


Figure 3. The SiPLS models for TBs (**a**,**b**), Caf (**c**,**d**), and EGCG (**e**,**f**) prediction. The green areas represent the optimal subintervals selected for predictive modeling.

3.3. Fluorescence Spectral and Chromatic Characteristics of SSDT Brew

3.3.1. The Distinctive Color and Fluorescence Signal of SSDT Brew

To establish a multidimensional fingerprint of SSDT, the fluorescence spectroscopy technique was employed to investigate the SSDT brew extensively, and its CIELab parameter was simultaneously recorded. Finally, 21 JJ and 23 KZ were subjected to fluorescence spectroscopy analysis after removing the samples with outliers based on the NIRS results. It was found that the SSDT brews were moderately high in lightness (L*), red (a*), and yellow (b*), and congregated together in the yellow–red zone in the CIE1931 xyY color space (x = 0.41 to 0.54 and y = 0.40 to 0.45) (Figure 4a). Moreover, the fluorescence EEM of the SSDT brew was subjected to PARAFAC, and four components were selected considering the core consistency [13]. Among them, component 1 showed the maximum fluorescence intensities at excitation wavelength (λex) 305 nm and emission wavelength (λem) 380 nm, while component 2 presented peak intensities at λex 265 nm and λem 315 nm. As for component 3, its peak intensities appeared at λex 335 nm and λem 445 nm, but the maximum intensities of component 4 were at λex 300 nm and λem 375 nm (Figure 4b). The fluorescence signal of the SSDT brew was mainly caused by the aromatic compounds, amino acid residues, polyphenols, vitamins, chlorophyll, and some other polymerization products, with fluorescence characteristics [39]. Specifically, based on the previous publications, the fluorescence signal derived from λex 265/270 nm and λem 310–315 nm could be tentatively assigned to catechin and its related compounds [40], suggesting that component 2 was primarily contributed by these chemicals. Notably, although the content of catechin-related compounds in SSDT is strikingly lower than that of other tea types, they may still represent characteristic chemicals of SSDT.

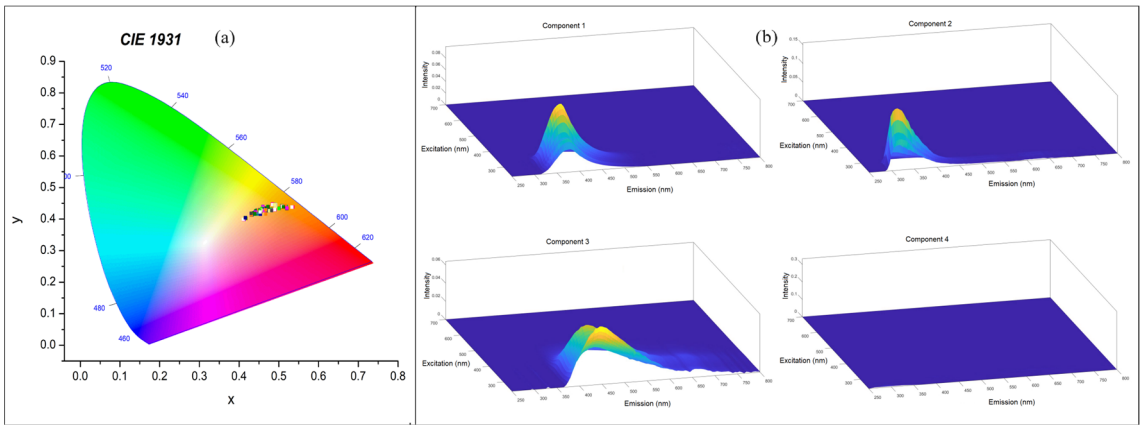


Figure 4. Fluorescence EEM and chromatic characteristics of SSDT brew: (**a**) CIE 1931 color space chromaticity diagram in the (*x*, *y*) coordinate system for SSDT; (**b**) fluorescence characteristics of SSDT brew based on PARAFAC, with 4 components. The *x*-axis in each component diagram represents the emission wavelength, the *y*-axis represents the excitation wavelength, and the *z*-axis represents the fluorescence intensity; the yellow color indicates the high fluorescence intensity, and the blue color indicates the low fluorescence intensity; the excitation wavelength at which the main fluorescence peak with the highest relative fluorescence intensity is located is the optimal excitation wavelength.

3.3.2. Identification of KZ and JJ

Principal component analysis (PCA), an unsupervised approach, and XGBDA, a supervised approach, were utilized to identify KZ and JJ based on their brew fluorescence EEM. As a result, in the PCA shown in Figure 5a, the first three principal components explained 97.78% of the total variance for the samples. The score plots of KZ and JJ showed a strong overlap, indicating that the KZ and JJ brew cannot be well distinguished by PCA. Conversely, as shown in Figure 5b, XGBDA exhibited excellent performance in separating KZ and JJ, with 100% sensitivity, 100% specificity, and 100% accuracy. This implies

that a supervised approach may be more appropriate for the SSDT brew fluorescence EEM analysis.

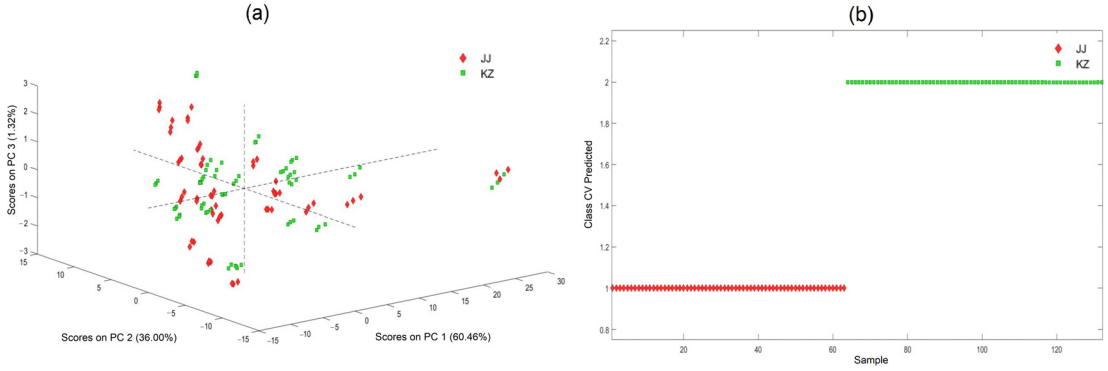


Figure 5. Classification of KZ and JJ based on the fluorescence EEM of tea brew. The red block represents JJ, and the green block represents KZ: (**a**) scores for KZ and JJ samples from PCA of fluorescence EEM data; (**b**) class cross-validation (CV) predicted most probable for KZ and JJ from XGBDA analysis of EEM data; no sample was misclassified.

4. Discussion

SSDT is typically made from the mature leaves and branches of tea plants. Its raw materials for production are among the oldest China's dark teas, resulting in lower levels of most common quality chemicals, especially TPs and AAs. Catechin is the primary constituent of TPs. Its oxidation products, i.e., TFs, TRs, and TBs, contribute significantly to the dark tea color. Here, the chemical CV differed dramatically among the SSDT samples, suggesting a potential variation in sensory qualities. Moreover, the lowest CV was discovered in TBs, and the CV of TFs, TRs, and TBs gradually decreased, but that of catechin monomers was higher, implying that TBs were stable among SSDT samples. TBs may be a characteristic compound reflecting the deep fermentation of phenolic compounds during SSDT production, and its substantial generation in this process may endow SSDT with a much stronger hypolipidemic effect [1].

Although TPs showed a lower level in SSDT due to the older or more mature raw materials, it still seems to be the critical compound in SSDT that seriously affected its NIR spectral fingerprint, since certain unique spectral bands highly related to polyphenols and catechins, such as the bands at 5780 cm^{-1} and 4500–4000 cm^{-1}, which were detected in this work [31,33]. Moreover, these chemicals also contributed significantly to the fluorescence spectral characteristics of the SSDT brew.

In addition, the NIR spectra of KZ and JJ presented a high similarity, possibly due to their identical production process and raw materials. The higher average spectral value of KZ may be significantly associated with its higher chemical levels than that of JJ. NIRS is a mature technology used in tea-related research, but its performance seems to vary, depending on tea types [6,9,17]. NIRS was able to accurately predict the levels of caffeine and catechins in Pu'erh ripe tea [8], indicating its appropriateness for quantifying such chemicals in dark tea. This was also confirmed in this work, as caffeine, EGCG, and TBs were excellently predicted by the NIRS-SiPLS models. Furthermore, efforts have been made to establish relationships or prediction models between chemicals, sensory quality, and NIRS. The sensory quality attributes of black tea have been successfully evaluated by NIRS [41], but similar analyses in SSDT are yet to be conducted.

Generally, consumers cannot distinguish KZ and JJ based on their dried tea appearance, which also could not be accurately identified by NIRS and the chemical analysis in this work. However, fluorescence EEM derived from the SSDT brew in combination with XGBDA exhibited excellent performance in their identification. The fluorescence EEM method is a 3D fluorescence technology that can provide complete information about the fluorescence signal of the SSDT brew, coupled with XGBDA, which showed an advantage

in discriminating highly similar samples. It has also been successfully applied to identify the vintage and geographical origin of wine [15], so this method could potentially be widely applied to the study of tea brew in the future.

Overall, this work presents the first comprehensive exploration of the spectral characteristics of dried SSDT and its brew, but some limitations still exist; for example, more samples with typical SSDT quality are still needed to improve the models when applied to the practical production and quality control of SSDT.

5. Conclusions

SSDT is a popular health drink with distinctive sensory quality. Dried SSDT has NIR spectral characteristics of relatively flat spectral curves at 10,000–8500 cm^{-1}, with a small absorption peak at 8334 cm^{-1} and more absorption peaks with higher absorbance values in the 8500–4000 cm^{-1} band. The main types of SSDT, i.e., KZ and JJ, showed very similar NIR spectra and cannot be distinguished by NIRS. TBs seem to be the most stable chemical compound in SSDT. The SiPLS-NIRS model established with the SNV + 2nd Der pretreatment presented excellent performance to predict TBs, Caf, and EGCG (RMSEC = 0.15, 0.12, and 0.02; RMSEP= 0.20, 0.09, and 0.03; Rc $\geq$ 0.92; and Rp $\geq$ 0.90). Simultaneously, fluorescence EEM combined with PARAFAC showed the fluorescence spectral characteristics of SSDT brew, with the maximum fluorescence intensities of components 1, 2, 3, and 4 at λex 305 nm/λem 380 nm, λex 265 nm/λem 315 nm, λex 335 nm/λem 445 nm, and λex 300 nm/λem 375 nm, respectively, while in combination with XGBDA, it effectively separated KZ and JJ with 100% sensitivity, 100% specificity, and 100% accuracy. The results can serve as a reference for the production, classification, and quality control of SSDT.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/agronomy14071582/s1, Figure S1: Sichuan south-road dark tea: (a) Kangzhuan; (b) Jinjian.; Table S1: Chemical details in the calibration and prediction sets; Table S2: Effects of different pretreatment methods on the PLS models; Table S3: Performance of SiPLS models established based on different combinations of spectral subintervals.

Author Contributions: Conceptualization, Y.Z.; methodology, Y.Z. and X.L.; data curation and visualization, Y.Z.; investigation, Y.Z., X.L. and D.H.; writing—original draft preparation, Y.Z.; writing—review and editing, Y.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by the "Disciplinary Construction Support Program" of Sichuan Agricultural University (1921993352).

Data Availability Statement: The original contributions presented in this study are included in the article/Supplementary Material. Further inquiries can be directed to the corresponding author.

Acknowledgments: The authors of this manuscript thank Susan E.P. Bastian and the School of Agriculture, Food and Wine, the University of Adelaide, for assistance with data collection.

Conflicts of Interest: The authors declare no conflicts of interest.

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